Muhammad Muizz Zafar

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Software Engineer | ETL Pipelines | Data Platforms | Cloud Infrastructure

Skills

Programming Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS, AWS CDK, CloudFormation, Docker, Splunk, CloudWatch, Athena, Apache Spark, Amazon

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - March 2025

Work experience

ETL Verification Developer — May 2024 - Present
Elections Canada — Ottawa, ON

- Built automated **Python ETL validation pipelines** and **Azure DevOps CI/CD workflows** to verify production data movement across **20M + records**, across multiple environments reducing manual validation effort.
- Used **Apache Spark** to profile, transform, and validate datasets of up to **5+ GB** during pipeline verification, accelerating bulk data checks and anomaly detection across staging and production environments.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to surface pipeline health, operational metrics, validation trends, and data quality issues for directorate and senior management.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of **production data infrastructure** across shared environments.
- Collaborated in an Agile cross-functional team with cloud engineers and DevOps teams to improve **distributed data workflows**, release quality, and production reliability.

Software Developer — September 2024 - December 2025
Kinaxis Inc. — Ottawa, ON

- Developed an automation framework to convert **35000** legacy KUnit tests written as sql queries to **JUnit** in **Java** by adding serilization modules at the start and end of each backend algorithm to capture the inputs and outputs for each test.
- Implemented **Java services** supporting scope control to detect incompatiblities between different features of the Rapid Response product which was used by customers and reduced L3 escalations by customers.
- Designed automated queries in SQL to detect **50+** known behavioral differences across different algorithms between the legacy and modular codebase.
- Used **Javascript** to write automated workflows within the rapid response product to run the known difference queries and visualize the known differences for directors, VP, and customers to see the differences between legacy and modular codebase

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Built **Docker** containers and automated event ingestion into **PostgreSQL** using **Python** and **GraphQl**, reducing testing time and improving deployment readiness.